

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

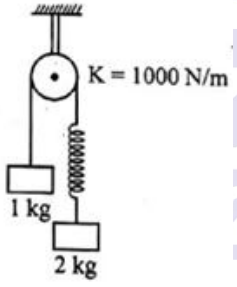
Sec: SR STAR - A & APEX, ALL IITS **DT: 29/03/2026**
Time: 3 Hrs **GTM - 08** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. In the arrangement shown in the figure pulley is light and smooth. The extension in the spring is ($g=10\ m/s^2$)



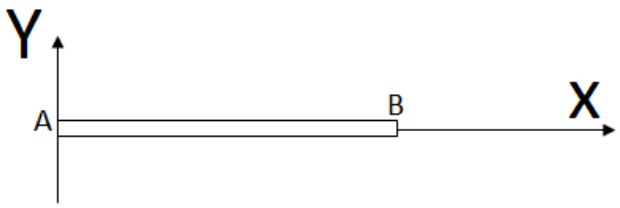
A) 1.33 cm B) 1.cm C) 1.67 cm D) 2 cm

For More Join: @IITJEE_Advanc

2. An object is projected with a speed 10 m/s at an angle of 30° with the horizontal. The object breaks down into n equal fragments during its motion. One fragment is found to strike the ground at a distance of $\sqrt{3}$ m from the point of projection in the same azimuthal plane, in which the object is projected. If the centre of mass of the remaining fragments strikes the ground simultaneously at distance of $7\sqrt{3}$ m from the point of projection, then the value of n is ($g=10\ m/s^2$)

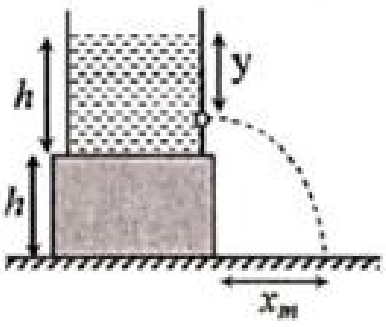
A) 2 B) 3 C) 4 D) 5

3. On the rod at point B a force $\vec{F}=\left(3\hat{i}+10\hat{j}+5\hat{k}\right)N$ is applied then the ratio of torque about other end 'A' and about y – axis is ____Nm



A) $\sqrt{5}$ B) $\sqrt{10}$ C) 5 D) $\sqrt{15}$

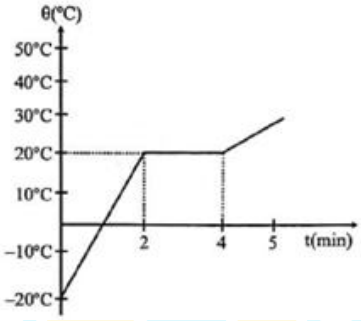
4. A tank is filled upto a height h with a liquid and is placed on a platform of height h from the ground. To get maximum range x_m a small hole is punched at distance of y from the free surface of the liquid. Then



A) $y=0.25h$ B) $y=0.5h$ C) $y=h$ D) $y=0.75h$

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5. Heat is supplied to 2kg of solid (initially at -20°C) at the constant rate of 5kJ/min. Temperature is plotted as a function of time as shown in the figure. Latent heat of fusion for solid is –



A) 10 kJ/kg B) 2.5 kJ/kg C) 5 kJ/kg D) 7.5 kJ/kg

6. A l cm long string fixed at both ends, sustains a standing wave such that the adjacent points on the string having displacement amplitude 1 mm (less than maximum amplitude) are separated by d cm. The string is oscillating in its third overtone then

A) $\frac{l}{d}=2$ B) $\frac{l}{d}=3$ C) $\frac{l}{d}=6$ D) $\frac{l}{d}=8$

7. Choose the INCORRECT STATEMENT.

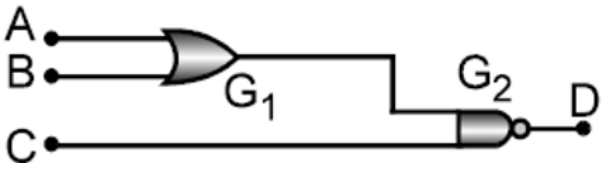
A) Gauss' law can be derived from Coulomb's law.

B) Gauss' law states that the net number of field lines crossing any closed surface in an outward direction is proportional to the net charge enclosed within the surface.

C) Coulomb's law can be derived from Gauss' law and symmetry.

D) According to Gauss' law, if a closed surface encloses no charge, then the electrostatic field must vanish everywhere on the surface.

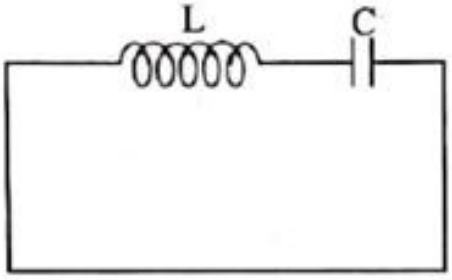
8. For the given combination of gates, if the logic states of inputs A, B, C are as follows $A=B=C=0$ and $A=B=1, C=0$ then the logic states of output D are



A) 0,0 B) 0,1 C) 1,0 D) 1,1

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9. In an LC circuit initially the capacitor has maximum charge q_0.Then the value of $\left(\frac{dI}{dt}\right)_{max}$ is



A) $\frac{q_0}{LC}$ B) $\frac{q_0}{\sqrt{LC}}$ C) $\frac{q_0}{LC}-1$ D) $\frac{q_0}{LC}+1$

10. The distance between object and screen is 96 cm. The ratio of length of two images formed by a convex lens placed between them is 4.84. The focal length f is nearly_

A) 22 cm B) 20.6 cm C) 34 cm D) 24.2 cm

11. In a screw gauge, there are 100 divisions in circular scale and each main scale division is of 1 mm. when there is no gap between the jaws, 97th division coincides with the main scale zero and zero of main scale is not visible. While measuring the diameter of a ball, the circular scale is between 3mm mark and 4 mm mark such that the 76th division of circular scale coincides with the reference line. Select the correct alternative.

A) The least count of the micrometer is 0.01 cm

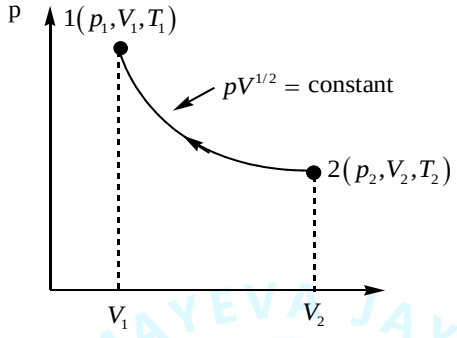
B) The zero error is -0.04 mm

C) The diameter of ball is 3.79 mm

D) The main scale reading is 4 mm

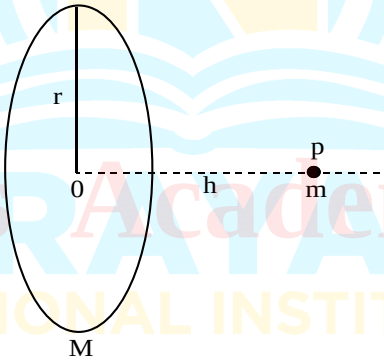
12. Consider a P-V diagram in which the path followed by one mole of perfect gas in a cylindrical container obeys law $PV^{\frac{1}{2}} = \text{constant}$ what is the ratio of the temperature $\frac{T_1}{T_2}$ if $V_2 = 2V_1$

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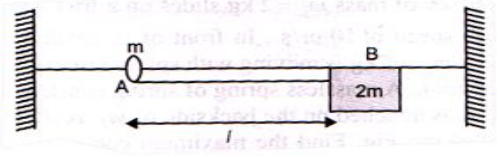
(A) $\frac{T_1}{T_2}=\frac{1}{2}$ (B) $\frac{T_1}{T_2}=\frac{2}{3}$ (C) $\frac{T_1}{T_2}=\frac{1}{\sqrt{2}}$ (D) $\frac{T_1}{T_2}=\frac{\sqrt{3}}{2}$

13. A mass m is placed at p at a distance h along the normal through the centre O' of a thin circular ring of mass M and radius r. Gravitational force on an object is F. If the mass is moved further away such that OP becomes 2h, by what factor the force of gravitation will decrease if h = r



(A) $F^{|}=\frac{4\sqrt{3}}{5}F$ (B) $F^{|}=\frac{4\sqrt{5}}{3\sqrt{2}}F$ (C) $F^{|}=\frac{4\sqrt{2}}{5\sqrt{5}}F$ (D) $F^{|}=\frac{5\sqrt{2}}{3\sqrt{3}}F$

14. A small ring A of mass m attached at an end of a light string of length *l*, the other end of which is tied to a small block B of mass 2m. The ring is free to move on a fixed smooth horizontal rod. Find the velocity of the ring when the string becomes vertical.



A) $\sqrt{5g\ell/3}$ B) $\sqrt{8g\ell/3}$ C) $\sqrt{g\ell/3}$ D) $\sqrt{4g\ell/3}$

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15. A solid sphere of mass m is lying at rest on a rough horizontal surface. The coefficient of friction between the ground and sphere is μ. The maximum value of F which is applied at centre horizontally, so that the sphere will not slip, is equal to

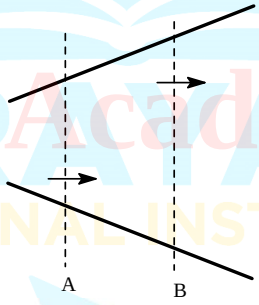
A) $\frac{7}{5}\mu mg$ B) $\frac{4}{7}\mu mg$ C) $\frac{5}{7}\mu mg$ D) $\frac{7}{2}\mu mg$

16. In YDSE, let A & B be two slits. Films of thickness t_A & t_B and refractive indices μ_A & μ_B are placed in front of A and B, respectively. If $\mu_A t_A = \mu_B t_B$, then central maxima will

A) not shift B) shift towards A, if $t_B > t_A$

C) shift towards B, if $t_B > t_A$ D) Shift towards A, if $t_B < t_A$

17. A non-viscous liquid is flowing through a non-uniform pipe from section A to section B as shown in the adjoining figure. Cross-sectional area of the pipe at A is less than that at B. Which of the following statement is **incorrect**?



A) The pressure at A is greater than that at B

B) Velocity at A is greater than that at B

C) Total energy per unit volume of the liquid is greater at A than at B

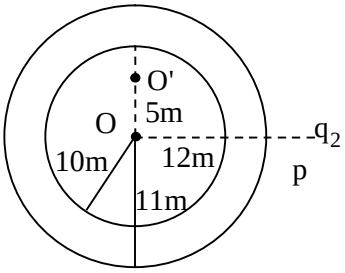
D) Axis of the pipe cannot be horizontal

18. A system consists of a uniform charged sphere of radius R and a surrounding medium filled by a charge with the volume density $\rho = \frac{\alpha}{r}$, where α is a positive constant and r is the distance from the centre of the sphere. The charge of the sphere for which the electric field intensity E outside the sphere is independent of r is

A) $\pi R^2 \alpha$ B) $4\pi R^2 \alpha$ C) $2\pi R^2 \alpha$ D) $3\pi R^2 \alpha / 4$

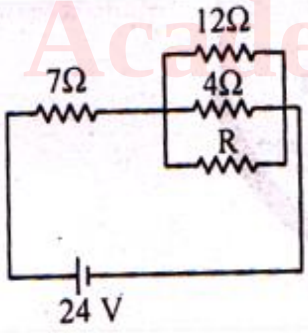
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19. A charge q_1 is placed at O, inside a hollow conducting sphere having inner and outer radii as 10m and 11m as shown. The force experienced by charge q_2 placed at point P is F_1 and force experienced by q_2 when q_1 is placed at O' is $F_2.(OO' = 5m, OP = 12m)$ Then $\frac{F_1}{F_2}$ is



A) 1 B) $\left(\frac{12}{13}\right)^2$ C) $\left(\frac{13}{12}\right)^2$ D) $\left(\frac{4}{3}\right)^2$

20. In the circuit shown, for what value of R, will the ideal battery transfer energy at the rate of 60 W



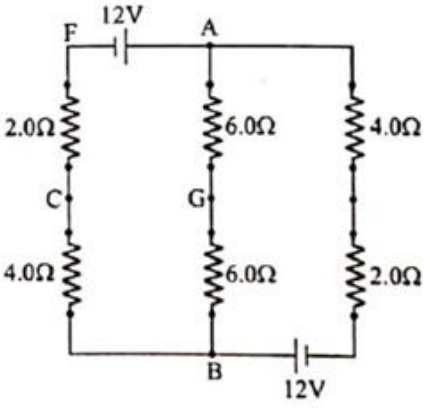
A) 12.5 B) 15.5 C)17.5 D) 19.5

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SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

21. In the network shown in the figure, the potential difference (in Volt) across points A and B is

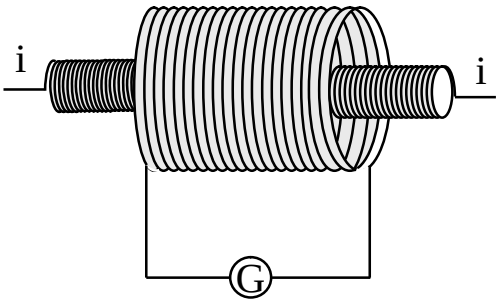


22. An ac source of angular frequency ω is connected across a resistance R and a capacitor C in series. The current registered is *i*. If now the angular frequency of source is changed to $\frac{\omega}{3}$ the current in the circuit is found to be halved. Then ratio of capacitive reactance to resistance at the original frequency is $\sqrt{\frac{n}{5}}$. Find value of n.

23. A light beam of power 12 watt and wavelength $6000A^o$ falls normally on a partially reflecting plate absorb 80% of light falling on it then force exerted on the plate is $4.8\lambda \times 10^{-n}$ Newton. Find the value of $\lambda + n$.

24. A long solenoid of radius 2 cm has 100 turns/cm and is surrounded by a 100-turn coil of radius 4 cm having a total resistance of 20Ω The coil is connected to a galvanometer as shown in figure. If the current in the solenoid is changed from 5A in one direction to 5 A in the opposite direction. The charge which flows through the galvanometer is $n\times10^{-4}C$. Then find value of n [Take $\pi^2 = 10$]

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25. A very large number of particles of same mass m are kept on positive x- axis at horizontal distances of 1metre, 2metre, 4metre, 8metre and so on from origin (0,0) point. The total gravitational potential at this point (origin) is $[-P\times m\times G]$ where G is universal gravitational constant and P is an integer. Find the value of P.

CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Match the following

Column – I (Treatment of reactant with excess water at room temperature)	**Column – II (Reaction Characteristics)**
1. $XeF_4 \xrightarrow{H_2O}$ 2. $XeF_6 \xrightarrow{H_2O}$ 3. $SF_6 \xrightarrow{H_2O}$ 4. $N_2O_4 \xrightarrow{H_2O}$	(P) $XeOF_4$ is one of the product (Q) One of the product is oxygen gas (R) Complete hydrolysis (S) Inert towards hydrolysis (T) One of the product is monobasic oxyacid

A) 1→ P, Q, R; 2 → P; 3→ S; 4 → P, R, T

B) 1→ P; 2 → P,Q; 3→ S; 4 → S, T

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C) 1→ Q; 2 → S; 3→ R; 4 → T

D) 1→ Q,R; 2 → P; 3→ S; 4 → R, T

27. Which combinations of quantum numbers, n, l, m and s for the electron in an atom does not provide a permissible solution of the wave equation?

A) $3,2,1,+\frac{1}{2}$ B) $3,1,1,-\frac{1}{2}$

C) $3,3,1,-\frac{1}{2}$ D) $3,2,-2,+\frac{1}{2}$

28. The correct statement about ICl_5 and ICl_4^- is

A) ICl_5 is square pyramidal and ICl_4^- is tetrahedral

B) ICl_5 is square pyramidal and ICl_4^- is square planar

C) Both are isostructural

D) ICl_5 is trigonal bipyramidal and ICl_4^- is tetrahedral

29. Among the following molecules/ ions, $C_2^{2-}, N_2^{2-}, O_2^{2-}, O_2$ which one is diamagnetic and has the shortest bond length?

A) C_2^{2-} B) O_2 C) O_2^{2-} D) N_2^{2-}

30. How many are example of narcotic analgesic?

(1) Aspirin (2) Paracetamol (3) Morphine

(4) Codeine (5) Heroin

A) 2 B) 3 C) 4 D) 5

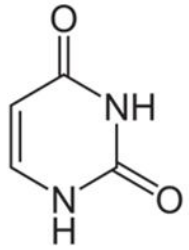
31. π_1, π_2, π_3 and π_4 atm are the osmotic pressures of 5% (mass/volume) solutions of urea, fructose, sucrose and KCl respectively at certain temperature. The correct order of their magnitudes is:

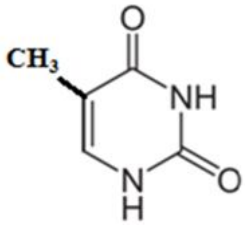
A) $\pi_1 > \pi_4 > \pi_2 > \pi_3$ B) $\pi_1 > \pi_4 > \pi_2 > \pi_3$

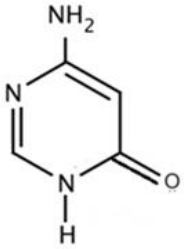
C) $\pi_4 > \pi_1 > \pi_2 > \pi_3$ D) $\pi_4 > \pi_1 > \pi_3 > \pi_2$

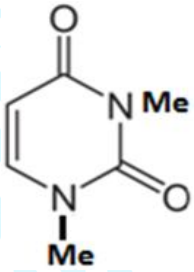
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32. Among the following compounds, which one is found in RNA?

A) 

B) 

C) 

D) 

33. A certain engine which operates in a carnot cycle absorbs 3.347 kJ at 400ºC, how much work is done by the engine per cycle ? The temperature of sink is 100ºC.

A) 1.49 kJ B) 2.51 kJ C) 1.85 kJ D) 2.98 kJ

34. An aqueous solution of a substance gives a white to give solution which is precipitate on treatment with dill. HCl which dissolves on heating. When H_2S is passed through the hot acidic solution, a black precipitate is obtained. The substance is a

A) Hg_2^{2+} *salt* B) Cu^{2+} *salt* C) Ag^{+} *salt* D) Pb^{2+} *salt*

35. Match the pollutant(s) in Column I with the effect(s) in Column II.

Column I	**Column II**
(i) Oxides of sulphur	(a) Global warming
(ii) Nitrogen dioxide	(b) Damage to kidney
(iii) Carbon dioxide	(c) 'Blue baby' syndrome
(iv) Nitrate in drinking water	(d) Respiratory diseases
(v) Lead	(e) Red haze in traffic and congested areas

A). (i) → (d), (ii) → (e), (iii) → (a), (iv) → (c), (v) → (b)

B). (i) → (b), (ii) → (e), (iii) → (a), (iv) → (c), (v) → (d)

C). (i) → (b), (ii) → (c), (iii) → (a), (iv) → (e), (v) → (d)

D). (i) → (d), (ii) → (c), (iii) → (b), (iv) → (e), (v) → (a)

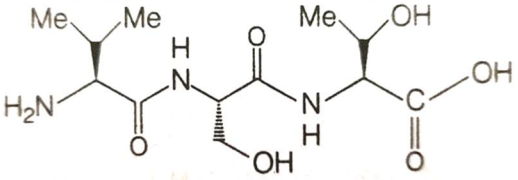
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36. Consider the hydrated ions of Ti^{2+}, V^{2+}, Ti^{3+} and Sc^{3+}. The correct order of their spin-only magnetic moments is

A) $Sc^{3+} < Ti^{3+} < Ti^{2+} < V^{2+}$ B) $Sc^{3+} < Ti^{3+} < V^{2+} < Ti^{2+}$

C) $Ti^{3+} < Ti^{2+} < Sc^{3+} < V^{2+}$ D) $V^{2+} < Ti^{2+} < Ti^{3+} < Sc^{3+}$

37. The correct sequence of amino acids present in the tripeptide given below is



A) Thr –Ser – Leu B) Leu – Ser- Thr

C) Val – Ser – Thr D) Thr – Ser – Val

38. Homoleptic octahedral complexes of a metal ion 'M^{3+} with three monodentate ligands L_1 ,L_2 and L_3 absorb wavelength in the region of green, blue and red respectively. The increasing order of the ligand strength is

A) $L_1 < L_2 < L_3$ B) $L_2 < L_1 < L_3$ C) $L_3 < L_1 < L_2$ D) $L_3 < L_2 < L_1$

39. Two liquids A and B have $P_A^0 : P_B^0 = 1:3$ at a certain temperature. If the mole fraction ratio of $x_A : x_B = 1:3$, the mole fraction of A in vapour in equilibrium with the solution at a given temperature is –

A) 0.1 B) 0.2 C) 0.5 D) 1.0

40. Reduction potentials of some ions are given below. Arrange them in decreasing order of oxidizing power?

Ion	ClO_4^-	IO_4^-	BrO_4^-
Reduction potential E^0 / V	$E^0 = 1.19V$	$E^0 = 1.65V$	$E^0 = 1.74V$

A) $IO_4^- > BrO_4^- > ClO_4^-$ B) $BrO_4^- > IO_4^- > ClO_4^-$

C) $BrO_4^- > ClO_4^- > IO_4^-$ D) $ClO_4^- > IO_4^- > BrO_4^-$

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41. Which of the following amine has more pK_b value in gases state

A) Methanamine B) N, N – Diethylethanamine

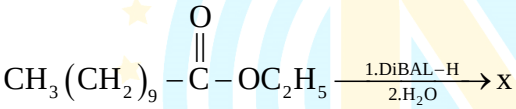
C) N – Ethylethanamine D) Ethanamine

42.

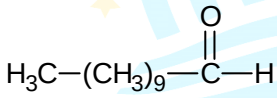
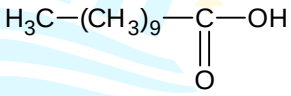
Compound	IUPAC name
a) o-chlorotoluene	2-Chlorotouluene
b) Benzyl chloride	Chlorophenyl methane
c) Allyl bromide	1-Bromo Prop-2-ene
d) tert-Butyl bromide	1-Bromo-2,2-dimethyl propane

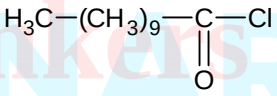
then incorrect IUPAC names are

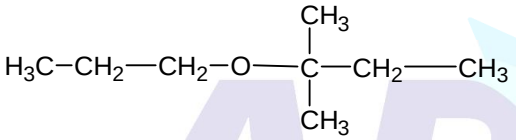
A) a, b B) b, c C) c, d D) a, d

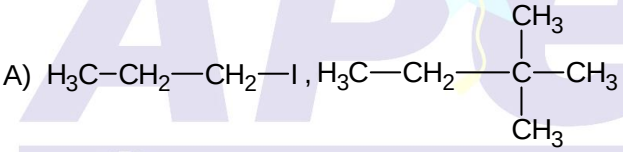
43. 

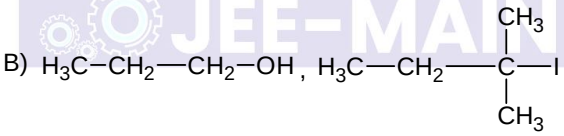
then formed product (x) is

A)  B) 

C)  D) $CH_3(CH_2)_9-CH_3$

44.  products (major) of the reaction are

A) 

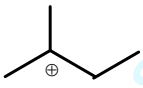
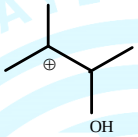
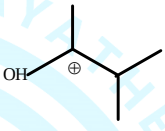
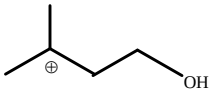
B) 

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C) $H_3C-CH_2-CH_2-OH$, $H_3C-CH_2-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-OH$

D) $H_3C-CH_2-CH_2-OH$, $H_3C-CH_2-\underset{\underset{CH_3}{|}}{CH}-CH_3$

45. Which of the following cation is most stable?

A)  B)  C)  D) 

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

46. The 'Spin Only' magnetic momentum for $\left[Ni(NH_3)_6\right]^{2+}$ in B.M is _____(nearest integervalue) .
(Given =Atomic number of Ni:28

47. How many of the following molecules are paramagnetic
N_2, NO, O_3, B_2, O_2, F_2.

48. The bond dissociation energy CH_4 and C_2H_6 are 360 and 620 kcal mol^{-1} respectively. The C-C bond energy is expected to be x. Then $\frac{x}{10} = ?$

49. During the estimation of nitrogen present in an organic compound (Containing three nitrogens per molecule) by kjeldahls method, the ammonia evolved from 0.5g of the compound in kjeldahls estimation of nitrogen, neutralized 10ml of $1MH_2SO_4$. Gram molecular weight (in gram) of the compound is

50. A direct current deposits 54 gr of silver (Atomic mass 108) during electrolysis. The weight of aluminium (Atomic mass 27) would be deposited from $AlCl_3$ solution by the same amount of current is 'x' gr, then the value of 10x is_____

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SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. If $y(\alpha)=\sqrt{1+2\tan\alpha(\tan\alpha+\sec\alpha)}$, $\alpha\in\left(0,\frac{\pi}{2}\right)$ then $\frac{dy}{d\alpha}$ at $\alpha=\frac{\pi}{6}$ is

A) 1 B) 2 C) 3 D) 4

52. The values of "a" to which one root of the equation $x^2-(a+1)x+a^2+a-8=0$ exceeds 2 and other is less than 2, are given by

A) $3<a<10$ B) $a>10$ C) $-2<a<3$ D) $a\le 2$

53. A tangent to the hyperbola $\frac{x^2}{4}-\frac{y^2}{2}=1$ meet x-axis at P and y –axis at Q, lines PR are QR are drawn such that OPRQ is a rectangle where O is origin then R lies on

A) $\frac{4}{x^2}+\frac{2}{y^2}=1$ B) $\frac{2}{x^2}-\frac{4}{y^2}=1$ C) $\frac{2}{x^2}+\frac{4}{y^2}=1$ D) $\frac{4}{x^2}-\frac{2}{y^2}=1$

54. The equation of the curve where it passing through origin and satisfying the differential equation $(1+x^2)\frac{dy}{dx}+2xy=4x^2$ is

A) $(1+x^2)y=x^3$ B) $3(1+x^2)y=2x^3$

C) $(1+x^2)y=3x^3$ D) $3(1+x^2)y=4x^3$

55. A seven digit number is formed by using all the digits 0, 1, 2, 3, 4, 8, 9 without repetition.Then the probability that it is divisible by 4 is

A) $\frac{53}{180}$ B) $\frac{53}{5(6!)}$ C) $\frac{53}{6(6!)}$ D) $\frac{53\times 4!}{5\times 6!}$

56. Let S be the set of real values of λ for which the function $f(x)=x^3-3(2\lambda-1)x^2+6\lambda x$ has exactly one local maximum and exactly one local minimum, then S can be

A) (0, 6) B) (1, 4) C) $(-\infty, 0)$ D) $(0, \infty)$

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57. A man has seven relative, 4 of them are ladies and 3 gentlemen, his wife also has 7 relative out of which 3 are ladies and 4 gentlemen. The number of ways in which they can invite a dinner party of 3 ladies and 3 gentlemen, so that there are 3 of the man's relative and 3 of the wife's relative, are

A) 144 B) 340 C) 484 D) 485

58. If $\vec{a}=\hat{i}-\hat{j}-\hat{k}$ and $\vec{b}=\lambda\hat{i}-3\hat{j}+\hat{k}$ and orthogonal projection of $\vec{b}$ on $\vec{a}$ is $\frac{4}{3}\left(\hat{i}-\hat{j}-\hat{k}\right)$ then λ is equal to:

A) 0 B) 2 C) 12 D) -1

59. Let $A=\left[a_{ij}\right]$ and $B=\left[b_{ij}\right]$ be two 3 x 3 real matrices such that $b_{ij}=(4)^{(i+j)-2}a_{ji}$, where i, j = 1, 2, 3. If the determinant of B is 256, then the determinant of A is

A) 4 B) 1/4 C) 1/256 D) 1/16

60. The ends of a line segment are P(1,3) and Q(1,1). R is a point on the line segment PQ such that R divides $\overline{PQ}$ internally in the ratio $1:\lambda$. If R is an interior point of the parabola $y^2=4x$ then

A) $\lambda\in(0,1)$ B) $\lambda\in\left(-\frac{3}{5},1\right)$

C) $\lambda\in\left(\frac{1}{2},\frac{3}{5}\right)$ D) $\lambda\in\left(-\frac{3}{5},\frac{3}{5}\right)$

61. P is a variable point on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=2$ (a>b)whose foci are F_1 and F_2. The maximum area (in unit^2) of the $\Delta PFF'$ is

A) $2b\sqrt{a^2-b^2}$ B) $\sqrt{2}b\sqrt{a^2-b^2}$

C) $b\sqrt{a^2-b^2}$ D) $2a\sqrt{a^2-b^2}$

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62. The shortest distance between the line $\frac{x-3}{3}=\frac{y}{0}=\frac{z}{-4}$ and the y-axis is

A) $\frac{1}{5}$ B) 1 C) 0 D) $\frac{12}{5}$

63. Let $f(x)=\begin{cases}\sin x, x\neq n\pi \\ 2, x=n\pi, \text{where } n\in Z\end{cases}$ and $g(x)=\begin{cases}x^2+1, x\neq 2 \\ 3, x=2\end{cases}$. Then $\lim_{x\to 0} g(f(x))$ is

A) 0 B) 1 C) 3 D) 2

64. The ordinate x=a divides the area bounded by the x-axis, the curve $y=1+\frac{8}{x^2}$ and the ordinates x=2 and x=4 into equal parts. Then a is

A) 3 B) $\frac{7}{2}$ C) $2\sqrt{2}$ D) $\frac{5}{2}$

65. The most general values of x for which $\sin x+\cos x=\min_{a\in R}\{1, a^2-4a+6\}$ are given by

A) $2n\pi; n\in z$ B) $2n\pi+\frac{\pi}{2}; n\in z$

C) $n\pi+(-1)^n.\frac{\pi}{4}-\frac{\pi}{4}; n\in z$ D) $2n\pi-\frac{\pi}{2}; n\in z$

66. If $(1+x+x^2+x^3)^{10}=a_0+a_1x+a_2x^2+a_3x^3+.....+a_{30}x^{30}$ then which of the following is true?

A) $a_2-a_4+a_6-.....-a_{28}+a_{30}=1$

B) $a_1-a_3+a_5-a_7+.....-a_{27}+a_{29}=0$

C) $a_0-a_1+a_2-a_3+.....+a_{30}=2^{19}$

D) Both (A) and (B)

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67. In a sequence of 21 terms the first 11 terms are in A.P. with common difference 2 and the last 11 terms are in G.P. with common ratio 2. If the middle term of the A.P. is equal to the middle term of the G.P., then the middle term of the entire sequence is

A) $\frac{-10}{31}$ B) $\frac{-630}{31}$ C) $\frac{-32}{31}$ D) $\frac{-320}{31}$

68. The chord of contact of the pair of tangents to the circle $x^2+y^2=1$ drawn from any point on the line $2x+y=4$ always passes through the point

A) $\left(\frac{1}{2},\frac{1}{4}\right)$ B) $\left(\frac{1}{4},\frac{1}{2}\right)$ C) $\left(1,\frac{1}{2}\right)$ D) $\left(\frac{1}{2},1\right)$

69. $\int \frac{x+x^{2/3}+x^{1/6}}{x\left(1+x^{1/3}\right)}dx =$

A) $\frac{3}{2}x^{2/3}+\tan^{-1}\left(x^{1/6}\right)+c$ B) $\frac{2}{3}x^{2/3}+6\tan^{-1}\left(x^{1/6}\right)+c$

C) $\frac{3}{2}x^{2/3}-6\tan^{-1}\left(x^{1/6}\right)+c$ D) $\frac{3}{2}x^{2/3}+6\tan^{-1}\left(x^{1/6}\right)+c$

70. If $\int_0^x f(t)dt = x+\int_x^1 tf(t)\,dt$, then the value of $f(1)$ is

A) $\frac{1}{2}$ B) 0 C) 1 D) $-\frac{1}{2}$

SECTION – II
(Numerical Value Answer Type)

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71. The system of linear equations $x_1+2x_2+3x_3=6$, $x_1+3x_2+5x_3=9$, $2x_1+5x_2+ax_3=b$ is consistent has infinite number of solutions, then the value of $a+b$ is

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72. For two data sets each of size 5. The variances are given by 4 and 5 and the corresponding means are given to be 2 and 4 respectively. If the variance of combined data is k then 6k equal to

73. Let A and B be two finite sets with m and n elements, respectively. The total number of subsets of the set A is 56 more than the total number of subsets of B. Then, the distance of the point $P(m, n)$ from the point Q (-2, -3) is

74. If $\sin x = \frac{-3}{5}$, where $\pi < x < \frac{3\pi}{2}$, then $80(\tan^2 x - \cos x)$ is equal to

75. If $z = 2 + 3i$, then $z^5 + (\bar{z})^5$ is equal to

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